CAPSTONE PROJECT

FAKE NEWS DETECTION BY MACHINE LEARNING

PRESENTED BY

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OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

Today, people get a lot of news from the internet and socail media. But some news is fake and can mislead people or create panic.

How can we check whether a news article is FAKE or REAL using a computer ?

PROPOSED SOLUTION

- I made a system that reads a news article and tells whether it is REAL or FAKE.
- I used:-
- A dataset that has many real and fake news examples.
- Machine Learning Learning to train a model on that data.
- TF-IDF to convert the news text into numbers.
- A classifier to detect fake news.

Technology used:-

Python
Pandas,Scikit-learn - library for ML
TF-IDF Vectorizer - For text processing
Passive Aggressivve Classifier - ML model used
Jupyter Notebook - Devlopment environment

SYSTEM APPROACH

Data Collection:

using a .csv file with labeled new whether it is fake or real

Preprocessing:

Removed unnecessary text

Converted text into numbers using TF-IDF

Model Training:

used 75% of data to train and 25% to test.

Model learned from examples

Prediction:

The model gives a prdiction: Fake or Real

ALGORITHM & DEPLOYMENT

- Algorithm Selection:
  - We used PassiveAggressiveClassifier from Scikit learn
  - It learns quickly and works well with text data.
- Data Input:
  - News Article Text
- Training Process:
  - The model was trained using labeled news data. Text was converted into numbers using TF-IDF, and a PassiveAggressiveClassifier was to learn patterns that separate Fake or Real news.
- Prediction Process:
  - New user input is transformed using TD-IDF and passed to the trained model. The model then predicts whether the news is Fake or Real. Real-time inputs can be taken through user text input.

RESULT

Accuracy Achieved: Around ~98%

The model gives correct prediction most of the time

It works well even with new, unseen news.

```
[12]: print("Model accracy:",round(acc*100,2),"%")

      Model accracy: 98.12 %
```

```
Enter a news(for exiting the program please enter 'exit'): AI will destory world !
Prdiction: FAKE
Enter a news(for exiting the program please enter 'exit'): Indian Prime Minister is Good!
Prdiction: REAL
Enter a news(for exiting the program please enter 'exit'): exit
```

CONCLUSION

Our System can help detect fake news automatically.

It uses macine Learning and real news data.

This tool can be useful in schools, colleges and social media platforms.

FUTURE SCOPE

Create a mobile or web app for real users.

Use Deep Learning for better accuracy.

Add more Languages and news sources.

Integrate it into social media for live checks.

REFERENCES

Scikit Learn offical documentation

Kaggle Dataset: news_combined.csv

Online tutorial for TF-IDF and PassiveAggressiveClassifier

LINKS

Video Link:- https://drive.google.com/file/d/1gZG-czGoPZQWnDMEppLnW4CokLh_7px-/view?usp=sharing

Github Link:- https://github.com/SinghBappi/NEWZ_AI-ML-Project

LinkedIn Post Link:- https://www.linkedin.com/posts/bappi-singh-13a15a33b_ai-machinelearning-fakenewsdetection-activity-7350214129116217344-lOxT?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFVGFT4Bv4f_Rebye2FBbgS7mURboFtvBcA

Thank you